

19-7-2026

التاريخ:

الموضوع:

Decision tree classification Algorithm

⇒ how to calculate purity?

↳ Entropy

↳ gini impurity $- (2)H = (1,2)IT$

⇒ how the Features are selected? to be splitted next

↳ information gain

* Decision tree is to split the data set into smaller subsets that are pure "containing only one class"

↳ identifies features that provide the Best separation between classes

* Entropy

⇒ $H(s) = -p^+ \log_2(p^+) - p^- \log_2(p^-)$

$p \rightarrow$ Probability

if Entropy = 0 ⇒ Pure Node

if data is 50/50 mixed

↳ Entropy increases towards

1 "maximum impurity"

⇒ long time complexity

⇒ used for more balanced models but takes time

* gini impurity

$G.I = 1 - \sum_{i=1}^n (p_i)^2$

if gini index = 0 ⇒ Pure Node

⇒ the split is chosen based on which feature results in the lowest gini score for the child nodes

⇒ if gini = 0.5 ⇒ impurity

⇒ faster calculations

* Information Gain (IG)

↳ also measures how much information we gain about target variable after splitting the data by a specific feature

$$\Rightarrow IG(s, f) = H(s) - \sum_{v \in \text{Vals}} \frac{|S_v|}{|S|} H(S_v)$$

entropy of s (Root node in D.Tree) no. of Vals for the child node entropy of child branches
 ↳ total num. of Vals - child nodes

⇒ based on $IG(s, f)$ Value we determine which ~~the~~ feature we start to split

$$\text{ex } \text{Gain}(s, f_1) = 0,049 \quad ; \quad \text{Gain}(s, f_2) = 0,051$$

$$\therefore \text{Gain}(s, f_2) > \text{Gain}(s, f_1)$$

⇒ then we start with f_2

* the highest information gain

* ~~~~~

* Classification Vs Regression Trees

I Decision Tree classifier

↳ Categorical data

⇒ Uses Entropy or Gini impurity to maximize (IG)

⇒ Final output or Leaf Node:

↳ Majority Vote ⇒ the class with the highest frequency in the leaf becomes the final prediction

2] Decision Tree Regressor :-

↳ continuous numerical data

⇒ uses Variance Reduction or Mean Square Error (MSE)

↳ Entropy and gini impurity are Not used here

⇒ leaf Node ⇒ Mean calculates the average of all numerical values in that specific leaf

* Splitting Logic "Handling input features"

⇒ Categorical Features "ex- Gender, city"

↳ perform a direct split for each category to evaluate which split yields the highest Information Gain

⇒ Continuous Features "ex- salary, weight"

↳ steps 1] Sort all numerical values in ascending order.

2] Threshold: calculates the average between every two consecutive numbers to create potential

Split Point "Threshold"

3] conditioning: Test a binary condition for every

threshold "ex $x \leq \text{Threshold}$ vs. $x > \text{Threshold}$ "

4] chooses the specific threshold that provides the maximum information gain (or maximum variance reduction)

* ROC "Receiver operating characteristic" and AUC curve:

↳ used in Binary classification problems and to measure the model's performance

Probabilities

في الخوارزمية اللوجستية Logistic Reg الفرجح من بيتا بالنتيجة مباشرة، لكنه يتنبأ باحتمالات في نطاق المسألة وفي طريقه الـ Threshold

كله لو الاحتمال اللي توقعه الفرجح اكبر من 0.5 يسا النتيجة 1 وكونه اقل من 0.5 نقبرها 0

⇒ Threshold: is a cut off point

↳ by default it's usually 0.5

we can change the threshold based on the problem

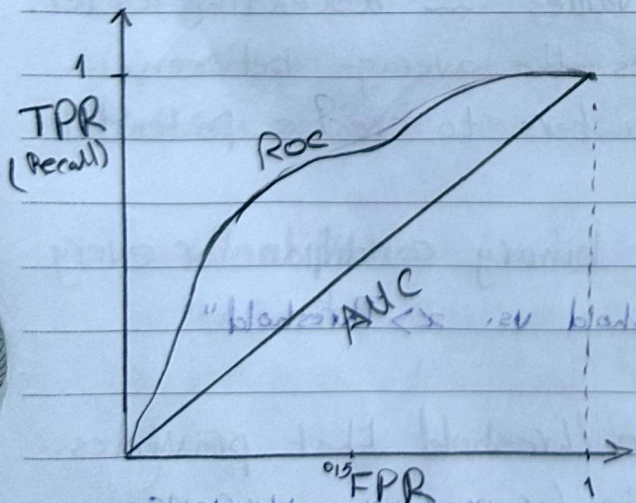
في حالة الطبيب الخطر الاكبر لو المريض مريض والفرجح يبي انو سليم (FN) والمريض يكون الـ FP هنا أكل عناه لو المريض سليم للموديل طلع صواب ويعمل التوقعات للأطباء ودة بيثبت على الـ threshold

* ROC Graph

"bigaw , probab" $TNR = \frac{TN}{FP+TN}$

$TNR = \frac{TN}{FP+TN}$

$FPR = \frac{FP}{FP+TN}$



* The bigger the AUC the better the model

* The Model's AUC must be higher than 0.5, if not then the model is dumb and doing Random guessing

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الموضوع:

* Confusion Matrix

consider the given confusion Matrix
For Binary classifier predicting the
Presence of a disease

	Predicted No	Predicted Yes
Actual No	45 TN	5 FP
Actual Yes	5 FN	95 TP

$$\Rightarrow \text{Accuracy} = \frac{TN+TP}{TN+FP+FN+TP}$$

$$= \frac{45+95}{150} = 93,33\%$$

$\Rightarrow$ Missclassification rate: overall how often is it wrong

$$\hookrightarrow \frac{FN+FP}{TN+FP+FN+TP} = \frac{5+5}{150} = 6,67\%$$

$\Rightarrow$ True Positive Rate: when actually Yes, how often predict Yes?

$$\hookrightarrow \text{Sensitivity} = \text{Recall}$$

$$\star \text{Recall} = \frac{TP}{\text{Actual Yes}} = \frac{95}{5+95} = 95\%$$

$\Rightarrow$ False Positive Rate: when actually No, how often it predict Yes?

$$\hookrightarrow \frac{FP}{\text{Actual No}} = \frac{5}{50} = 10\%$$

$\Rightarrow$ True Negative Rate: when actually No, how often predict No?

$$\hookrightarrow \text{Specificity} = \frac{TN}{\text{Actual No}} = \frac{45}{50} = 90\%$$

$\Rightarrow$ Precision: when it predicts yes, how often is it correct?

$$\hookrightarrow \frac{TP}{\text{Predicted Yes (TP+FP)}} = \frac{95}{95+5} = 95\%$$

$\Rightarrow$ Prevalence: How often does the yes condition actually occur in our sample

$$\hookrightarrow \frac{\text{Actual Yes}}{\text{total}} = \frac{100}{150} = 66,67\%$$